FINITE PRESENTATION RELATIVE TO R SURVIVES LOCALIZATION: M OF FINITE PRESENTATION OVER R_F BECOMES M_G OF FINITE PRESENTATION OVER R

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Editorial note. The last sentence of the proof as printed says that the displayed map is a presentation of M_f, while the construction is set up to present M_g. That slip is copied verbatim and is not corrected, since the lane delivers the author's own text and does not edit mathematics.

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Allowed standard prerequisites (editorial).

- The localization R_f is the quotient of the polynomial ring in one variable by fx_0 - 1, so a presentation over R_f can be rewritten as a presentation over R with one extra variable. used at proof lines 21023-21026, where R_f = R[x_0]/(fx_0 - 1) is recorded and the presentation is extended by the variables x_0 and x_{n+1}. Produces the ambient polynomial ring over R in which the enlarged presentation of the localized module is written.
- Elements and matrix entries of a finitely generated algebra over R_f can be lifted to the polynomial ring over R. used at proof lines 21027-21034, where g' and the entries h'_ij are chosen to map to g and to h_ij. Supplies the blocks of the lifted matrix, which is what makes the presentation a presentation over R rather than over R_f.
- A surjection of free modules with a given matrix, displayed with three blocks, presents the localized module once the second block encodes fx_0 - 1 and the third block encodes x_{n+1}g' - 1. used at proof lines 21037-21044, where the t by (s + 2t) matrix is described block by block. Verifies that the displayed map is onto M_g, which is the conclusion of the lemma.

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Lemma 0.2. *Let R be a ring, $f \in R$ an element, $R_f \to A$ is a finite type ring map, $g \in A$, and M an A-module. If M of finite presentation relative to R_f, then M_g is an A_g-module of finite presentation relative to R.*

Proof. Choose a presentation $R_f[x_1, \ldots, x_n] \to A$. We write $R_f = R[x_0]/(fx_0 - 1)$. Consider the presentation $R[x_0, x_1, \ldots, x_n, x_{n+1}] \to A_g$ which extends the given map, maps x_0 to the image of $1/f$, and maps x_{n+1} to $1/g$. Choose $g' \in R[x_0, x_1, \ldots, x_n]$ which maps to g (this is possible). Suppose that

$$R_f[x_1, \ldots, x_n]^{\oplus s} \to R_f[x_1, \ldots, x_n]^{\oplus t} \to M \to 0$$

is a presentation of M given by a matrix (h_{ij}). Pick $h'_{ij} \in R[x_0, x_1, \ldots, x_n]$ which map to h_{ij}. Then

$$R[x_0, x_1, \ldots, x_n, x_{n+1}]^{\oplus s+2t} \to R[x_0, x_1, \ldots, x_n, x_{n+1}]^{\oplus t} \to M_g \to 0$$

is a presentation of M_f. Here the $t \times (s + 2t)$ matrix defining the map has a first $t \times s$ block consisting of the matrix h'_{ij}, a second $t \times t$ block which is $(x_0 f -)I_t$, and a third block which is $(x_{n+1}g' - 1)I_t$. $\square$